

User-Centered Property Adjustment with Programmable Filament

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ABSTRACT

In today's society, where "one size fits all" does not work anymore, people's requirements for material living became more personalized. Despite the fact that there exist customized services for a variety of commodities, these services are generally expensive and such customization for physical artifacts is not accessible by average users. In this work, we follow the promise of Programmable Filament [21], a novel technique to enable end-users with low-cost Fused Deposition Modeling (FDM) 3D printers to equip multi-material printing capabilities at low investment. We discovered that this technique can be applied to mix mechanical properties of two different materials (e.g., tensility) that may affect user's comfort in 3D printed objects (e.g., personal optimal softness in sports gear) to meet individual needs.

We briefly introduce the process to 3D print programmable filaments using consumer-grade 3D printers in various composition ratios and fabricate 3D printed objects using an FDM printer blending multi-materials in the hot end, present a tensile testing experiment on sample objects made of such filaments to show its feasibility in producing new properties, and demonstrate two real-world applications that can be benefited from Programmable Filament utilization. We conclude with the remaining limitations and discussions about two potential structural improvements to improve printability of this filament for deployment.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**.

KEYWORDS

3D printing, programmable filament, multiple materials, tensility, tensile testing, user comfort

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1 INTRODUCTION

With the fast development of current society, people's demands on material living are not easily satisfied with the basic standard production of many consumer products; the requirements for custom material comfort and practicability are also worthy of attention. Unfortunately, most commodities are produced in factories with uniform standards [7], while different people might have different preferences to meet varying needs and individual tasks in a wide spectrum of daily lives, such as sports gears, toys, personal rehabilitation devices, and beyond. While there exist services to tailor-make personal products, they are often expensive and the process is not accessible to average users.

With the growing advances in personal fabrication, we see Programmable Filament[21] as a promising approach to tackle material and property customization for individuals. Programmable Filament is a novel technique that engages end-users with cheap FDM printers at home to create a new fabrication process. Apart from the previous world of personal fabrication that is often associated with the *shape customization*, it adds another level of *function/property customization* by introducing multi-material printing capabilities without any hardware updates. We applied this technique to print dual-material filaments with Thermoplastic Polyurethane (TPU), well-known for its flexibility and stretchability, and Polylactic Acid (PLA), which is the most common regular, rigid plastic material. We hypothesized that we can obtain the medium-grade property of mechanical characteristics that these two materials share but vary in degrees. For example, as the TPU can present a greater buckling behavior that PLA does not own, the composite material of the two may present would be less flexible in its bending capability than the pure TPU with the contribution of PLA. In the tensile testing experiment we conducted later, we discovered that this material composite of 60% TPU and 40% of PLA in ratio displays the medium tensility that is closer to TPU's tensility accounting for its contribution, which is the most visible property. The result implies that it would be feasible to proceed (1) to compose two materials in

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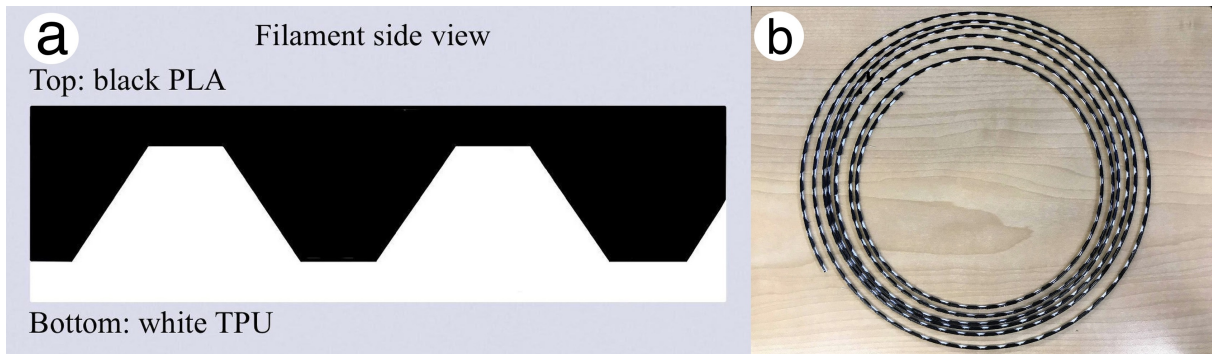


Figure 1: Details of the filament side view(a) and an exemplary printed filament with TPU & PLA materials(b).

varying ratios, as well as (2) to use various material compositions beyond TPU and PLA materials to explore a wider variety of other mechanical properties, proposing that Programmable Filament can be applied to many practical real-world applications for the benefit of end-users to adjust the level of desired properties to meet individual needs.

In the following sections, the concept and principles to produce Programmable Filament are briefly introduced first, then the basic design and printing procedures of Programmable Filament are described. A tensile test experiment is conducted for its experimental result to be analyzed. We show two real-world applications that can benefit from Programmable Filament. We conclude this work by discussing the remaining limitations and proposing two potential improvements to the filament segment structure.

2 RELATED WORK & BACKGROUND

Despite the recent growth of low-cost Fused Deposition Modeling (FDM) 3D printing, limited material choices have prevented non-experts from creating a wide spectrum of 3D printed objects with custom properties. The current mechanisms of low-cost printers produce objects in the layer by layer fashion, mostly using a single extruder and rigid plastic resulting in limited properties in final products. Structural arrangements of microstructures at various scales affect the material properties, existing works have tackled similar method [6, 11, 12, 16]. However, those approaches often require expert knowledge and training, which is not suitable for many potential end-users. Recent works thus have begun to utilize G-code, a set of machine commands informing slicing parameters (e.g., nozzle speed, printing temperature). Such work has achieved advanced material properties that are not possible using rigid plastic and consumer-grade machines, such as 3D printing of hairs and flexible textiles [4, 9, 14, 19, 20]. Compared to the latest 3D printing of multi-material solutions that may require expensive hardware, knowledge of chemical processes, and high-tech laboratory settings (e.g., [8, 17, 18, 23]), the use of new structure in filaments has showcased their practicality in many domains such as optoelectronics [10] or mission critical parts [15].

This project aims to demonstrate how dual-material filaments conducted by Programmable Filament[21] can show medium properties of two different materials. Programmable Filament is a novel technique that enables users to print multiple-material objects with

low-cost, single-nozzle FDM 3D printers without hardware upgrades. Numerous variables such as filament length, thickness, segment structure can be adjusted by manipulating the G-code. The overall workflow of this technique is to create a G-code file – print a long spiral with desired materials – use printed filaments as ready-to-use ones to generate 3D printing objects.

Specifically in this paper, we applied Programmable Filament in comprising dual-material filaments which are composed of the TPU and PLA materials, where the white TPU filament is located on the bottom and the black PLA filament is on the top (Figure 1). The contact surface of adjacent segments is diagonal to avoid nozzle collision during printing the second material on top [[21]]. The filament length, diameter, and angle among segments may differ depending on the needs of different experiments. In this work, we set the diameter to 1.6 millimeter and segment angle to 60 degrees.

3 APPROACH

When printing such filaments using the G-code generated from the schematics, users need to insert the one filament first in which a brim adhering to the printing surface to be generated to tightly locate the bottom parts of the lower half. Then trapezoid-shaped segments will be printed onto the brim. The 3D printer pauses after printing all white TPU segments and emits a beep sound signal for material exchange. At this point, users manually switch to a second filament and press the start button to resume the rest of the printing. The 3D printer then prints the same trapezoid-shaped segments upside down and finishes with a brim on top so that the outcome forms a near-cylindrical shape that resembles filaments used in 3D printing for actual object printing. Carefully removing the filament from the printing bed after completion of printing and getting rid of brims to post-process gives users a ready-to-print filament in new properties generated as shown in Figure 1. Often, printing with rigid plastic first is required to guarantee successful printing of mechanically distal materials together [22].

4 PROGRAMMABLE FILAMENT TENSILE ANALYSIS

4.1 Sample Preparation: Dog-Bone shaped Object

Dog-bone samples [3] are often produced for many mechanical testing. In this work, we prepared uniform sized dog-bone samples (Figure 2) according to type IV standards [3]. Each dog-bone sample takes approximately 3 meters of 3D printed filament to print. Due the fact that raster angle can affect object printing accuracy and performance, such as shape-recovery rate[5], we have set three different raster angles: 0 degree, 45 degrees, and 90 degrees. For each raster angle setting, samples are printed with TPU, PLA, and TPU-PLA filaments.

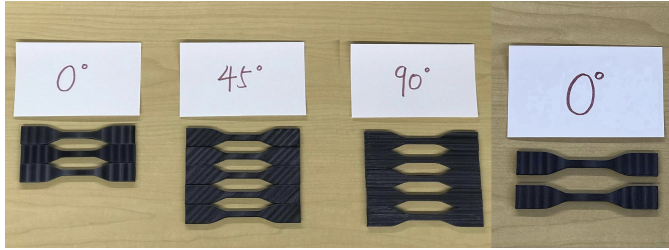


Figure 2: Five TPU-PLA dog-bone samples of each raster angle: 0 degree, 45 degrees, and 90 degrees.

In addition to the filament configuration we mentioned earlier (series configuration) for all samples, we also prepared TPU-PLA samples made of parallel configuration filament (Figure 3). Specifically, the TPU-PLA filaments with parallel configurations have a TPU-PLA ratio of 50:50, and the ones with series configurations have a TPU-PLA ratio of 60:40.

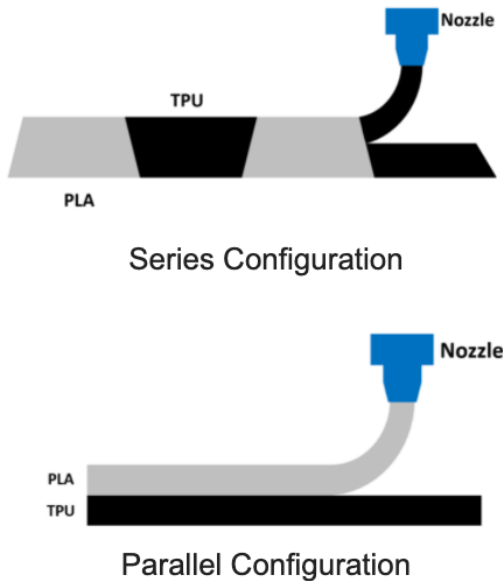


Figure 3: Parallel and series configuration of the TPU-PLA filament.

4.2 Tensile Testing Process

The tensile testing experiment is designed to explore the tensility of TPU, PLA, and the mixture of these two. Dog-bone samples are separated into three main categories by different raster angle: 0 degree, 45 degrees, and 90 degrees. At each raster angle, dog-bone samples are further organized according to their composition: TPU, PLA, and TPU-PLA mixture.

Dog-bone samples are subjected to uniaxial tensile loads in an Instron 5984 Floor Standing Universal Testing Machine [1], two ends of a sample are grasped by pinches, and the behaviors of the center are measured by seeing the force applied to break the connection as shown in Figure 4. The control variable is strain, and it is applied by the lab equipment with a moving speed of 0.0394 inch per second. The stress (pascal) must be applied in order to respond to the corresponding strain will be measured. At the end of this experiment, we would organize all experimental data and compare the stress-strain relationship of each group of dog-bone samples to draw conclusions in terms of tensility.

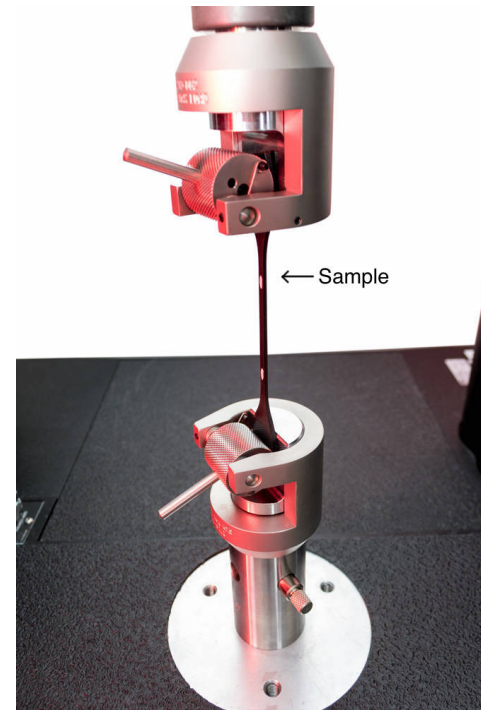


Figure 4: Two ends of a sample placed in Instron 5984 machine are grasped by pinches, and the behaviors of the center are measured by seeing the force applied to break the connection [1].

4.3 Results

The experimental data as shown in Figure 5 presents the stress-strain relationship of different groups of dog bone samples. The x-axis is strain, which is a quantity to describe how much the sample deforms under applied force. the y-axis is stress(pascal), which is the force applied to samples in order to respond to corresponding

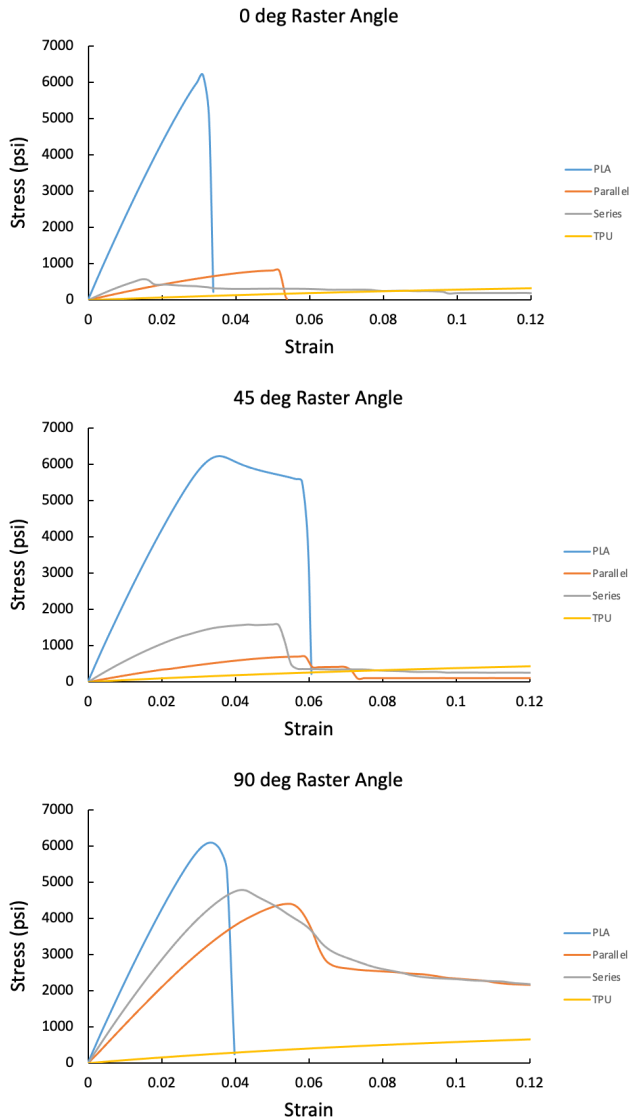


Figure 5: Tensile testing result of dog-bone samples made of TPU, PLA, and TPU-PLA mixture in all three raster angle circumstances.

strain [2]. Basic on strain, we compared the measured stress of each sample; the smaller the stress, the easier the sample to be stretched, and therefore this sample has a better tensility. The sample data provided in Figure 5 is the average value of all samples within the same material and raster angle group.

According to our experimental data, TPU-made dog-bone samples have the best tensility than others in all three raster angles circumstances. Overall, TPU-made dog-bone samples have the lowest stress-strain curves, which indicates that TPU requires the least stress in order to be deformed or stretched, compared to the other two materials. The PLA-made samples are in a contrary way, their

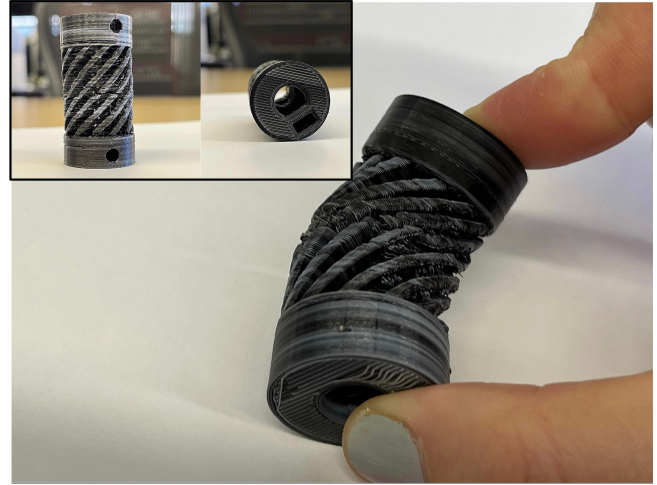


Figure 6: An exemplary printed flexible coupling with the TPU-PLA filament can be bent with hands while maintaining a certain firmness.

stress-strain curves are the highest among all samples, which indicates large stress needs to be performed in order for them to approach a relatively small strain. Dog-bone samples made of TPU-PLA mixture (parallel and series data) have stress-strain curves in the middle between TPU and PLA data in all three raster angle diagrams, which means the TPU-PLA mixture has better tensility than PLA and less tensility than TPU. Compared to the TPU-PLA samples with series configuration filament (composite of 50% TPU and 50% of PLA in ratio), the ones with parallel configuration filament (composite of 60% TPU and 40% of PLA in ratio) have stress-strain curves that are closer to TPU-made samples' curves.

The experimental result demonstrates that the composition of TPU and PLA displays the medium property of the two materials in terms of tensility, and this medium-grade property can be adjusted by varying ratios of materials. With this conclusion in mind, we can further apply this principle to real-world applications and future research in discovering more composition beyond TPU and PLA and a variety of other mechanical properties.

5 REAL-WORLD APPLICATIONS

After learning the properties of Programmable Filaments made of TPU and PLA, we looked at real-world applications and selected two 3D printing objects that can benefit from such properties.

5.1 Flexible Coupling

Flexible linkage [24] is an application that can be applied to a wide range of fields, particularly the field of engineering. This coupling has a straight hole through the central part to support and connect hardware pieces that can fit this hole (Figure 6). With the flexibility of adjusting multiple variables, e.g., length, diameters and hole shapes, this flexible coupling can well fit various hardware parts and be used in the end float coupling, as well as solve problems such as angular and offset alignments.

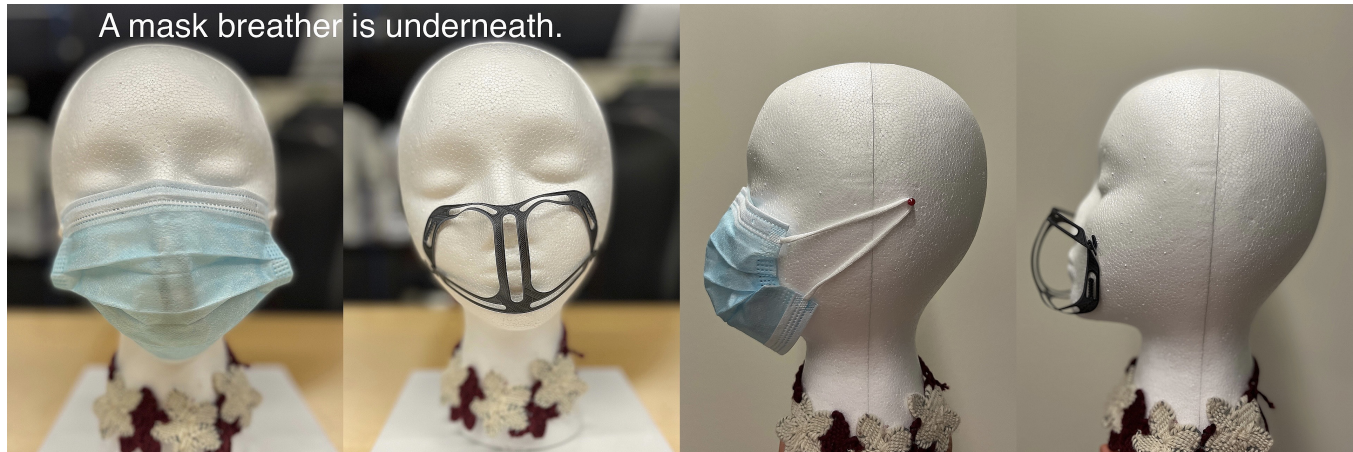


Figure 7: The Covid 19 mask breather can expand space between human faces and masks with personal comfort.

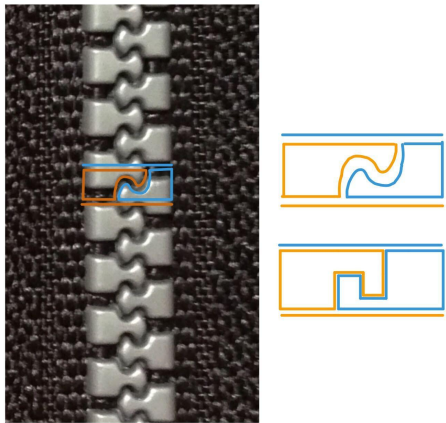


Figure 9: Two potential different zipper-shape structures to reinforce segments joints.



Figure 8: 3D printed Covid 19 mask breather with the TPU-PLA filament is printed flat with hooks to bend its shape.

In addition to shape adjustments, users may have demanded tensility for different scenarios. The TPU material offers flexibility due to its softness, while the PLA material is relatively firm and contributes to great support. A coupling consisting of the TPU-PLA filament can combine flexibility and firmness and exhibit an

average strength of the two materials. The exemplary printed flexible coupling with the TPU-PLA filament can be bent with hands while maintaining a certain firmness. Users can approach different tensility by adjusting the ratio of TPU and PLA in filaments.

5.2 Mask Breather

The COVID-19 respirator [13] is a flexible mask-shaped application that can expand the space between human faces and masks (Figure 7). It's printed flat with hooks to bend its shape (Figure 8). Besides of size, users can adjust the level of comfort by modifying its tensility. In order to obtain greater softness of this breather, users need to achieve better tensility by increasing the percentage of TPU in filaments. On the other hand, if users prefer more support in this breather, they may get less tensility by increasing the percentage of PLA in filaments instead.

6 LIMITATION & FUTURE WORK

6.1 Improving the Printability of Programmable Filaments by increasing the Binding Force

Even though the result has met our expectations, the actual manipulation of printing Programmable Filament is less than ideal. Compared to other ready-to-use filaments, our Programmable Filament design at the current stage connects different materials by the trapezoid-shaped structure (Figure 1), which is not the optimal connection structure to provide enough binding force. Due to the adhesion of the materials and the trapeze-like connection, filament breakage would occur when it meets pressure near the gear and friction inside of the filament tube. Given that the mechanical properties of TPU and PLA materials are non-modifiable, we can make adjustments in the programmable filament structure.

Two potential programmable filament structure modifications for improving the extruding process are proposed here. (1) Besides the top brim and bottom brim, add extra brims on both left and right sides to hold segments in place. (2) Changing current connection structure to zipper-shape to reinforce segment joints (Figure 9).

6.2 Materials and Mechanical Properties

From the experimental result of tensile test, we prove that the composition of PLA and TPU shows the medium-grade property of these two materials in terms of tensility, and we present two real-world applications that can benefit users by providing them the tensility-adjusting option. Our next approach would be proceeding to use various material compositions beyond TPU and PLA and explore a wider variety of other mechanical properties, so end-users can adjust more properties of real-world applications to fit individual needs.

7 CONCLUSION

In this work, we demonstrate that users can apply the Programmable Filament technique to produce various material compositions filaments, therefore adjust object properties to fit individual needs. We briefly introduced the Programmable Filament technique, conduct a tensile testing experiment to understand how tensility would be affected by mixing TPU and PLA materials, then present two real-world applications printed by the TPU-PLA filament that can benefit users by adjusting tensility. Finally, we discussed current limitation of Programmable Filament connection structure, two potential approaches to improve structure connectivity, and future steps.

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REFERENCES

- [1] 2022. 5984 Floor Standing Universal Testing Machine, 5980 Series. Retrieved Jan 13, 2022 from <https://www.instron.com/en/products/testing-systems/universal-testing-systems/high-force-universal-testing-systems/5980-series-floor-models>
- [2] A. F. C. Brown and N. D. G. Vincent. 1941. The Relationship between Stress and Strain in the Tensile Impact Test. *Proceedings of the Institution of Mechanical Engineers* 145, 1 (1941), 126–134. https://doi.org/10.1243/PIME_PROC_1941_145_029_02
- [3] ASTM D638-14. 2017. *Standard Test Method for Tensile Properties of Plastics*. ASTM International, West Conshohocken, PA. <https://doi.org/10.1520/D0638-14>
- [4] Jack Forman, Mustafa Doga Dogan, Hamilton Forsythe, and Hiroshi Ishii. 2020. DefeXtiles: 3D Printing Quasi-Woven Fabric via Under-Extrusion. In *Proceedings of the 33rd Annual ACM Symposium on User Interface Software and Technology* (Virtual Event, USA) (UIST '20). Association for Computing Machinery, New York, NY, USA, 1222–1233. <https://doi.org/10.1145/3379337.3415876>
- [5] Hassan Gonabadi, Yao Chen, Arti Yadav, and Steve Bull. 2021. Investigation of the effect of raster angle, build orientation, and infill density on the elastic response of 3D printed parts using finite element microstructural modeling and homogenization techniques. *Int J Adv Manuf Technol* 18 (Sept. 2021), 1485–1510. <https://doi.org/10.1007/s00170-021-07940-4>
- [6] Zvi Hashin and Shmuel Shtrikman. 1963. A variational approach to the theory of the elastic behaviour of multiphase materials. *Journal of the Mechanics and Physics of Solids* 11, 2 (1963), 127–140.
- [7] S. Jack Hu. 2013. Evolving Paradigms of Manufacturing: From Mass Production to Mass Customization and Personalization. *Procedia CIRP* 7 (2013), 3–8. <https://doi.org/10.1016/j.procir.2013.05.002> Forty Sixth CIRP Conference on Manufacturing Systems 2013.
- [8] Tyler Koslow. 2016. *Pattette2S*. <https://www.mosaicmfg.com/products/palette-2s>
- [9] Gierad Laput, Xiang 'Anthony' Chen, and Chris Harrison. 2015. 3D Printed Hair: Fused Deposition Modeling of Soft Strands, Fibers, and Bristles. In *Proceedings of the 28th Annual ACM Symposium on User Interface Software & Technology* (Charlotte, NC, USA) (UIST '15). ACM, New York, NY, USA, 593–597. <https://doi.org/10.1145/2807442.2807484>
- [10] Gabriel Loke, Rodger Yuan, Michael Rein, Tural Khudiyev, Yash Jain, John Joannopoulos, and Yoel Fink. 2019. Structured multimaterial filaments for 3D printing of optoelectronics. *Nature Communications* 10, 1 (Sept. 2019). <https://doi.org/10.1038/s41467-019-11986-0>
- [11] Jean-Claude Michel, Hervé Moulinec, and P Suquet. 1999. Effective properties of composite materials with periodic microstructure: a computational approach. *Computer methods in applied mechanics and engineering* 172, 1-4 (1999), 109–143.
- [12] Graeme W. Milton. 2002. The theory of composites. *The Theory of Composites*, by Graeme W. Milton, pp. 748. ISBN 0521781256. Cambridge, UK: Cambridge University Press, May 2002. (2002), 748.
- [13] Jo Ri. 2020. *Covid 19 mask breather: Remix, full flat 3D print or laser cut*. Retrieved January 10, 2022 from <https://www.thingiverse.com/thing:4599585>
- [14] H. Tanaka S. Koda. 2017. Direct G-code manipulation for 3D material weaving. In *Proceedings of SPIE Smart Structures and Materials + Nondestructive Evaluation and Health Monitoring*, Vol. 10167. <https://doi.org/10.1117/12.2261648>
- [15] Kubi Sertoglu. 2020. 3D Printing Industry: U.S. Army develops high strength multi-polymer filament for use with low-cost 3D printers. <https://3dprintingindustry.com/news/u-s-army-develops-high-strength-multi-polymer-filament-for-use-with-low-cost-3d-printers-171038/>. (Accessed on 03/01/2022).
- [16] Ole Sigmund. 1995. Tailoring materials with prescribed elastic properties. *Mechanics of Materials* 20, 4 (1995), 351 – 368. [https://doi.org/10.1016/0167-6636\(94\)00069-7](https://doi.org/10.1016/0167-6636(94)00069-7)
- [17] Pitchaya Sitthi-Amorn, Javier E. Ramos, Yuwang Wang, Joyce Kwan, Justin Lan, Wenshou Wang, and Wojciech Matusik. 2015. MultiFab: A Machine Vision Assisted Platform for Multi-Material 3D Printing. *ACM Trans. Graph.* 34, 4, Article 129 (jul 2015), 11 pages. <https://doi.org/10.1145/2766962>
- [18] Mark A. Skylar-Scott, Jochen Mueller, Claas W. Visser, and Jennifer A. Lewis. 2019. Voxlated soft matter via multimaterial multinozzle 3D printing. *Nature* 575, 7782 (Nov. 2019), 330–335. <https://doi.org/10.1038/s41586-019-1736-8>
- [19] Haruki Takahashi and Jeeun Kim. 2019. 3D Printed Fabric: Techniques for Design and 3D Weaving Programmable Textiles. In *Proceedings of the 32nd Annual ACM Symposium on User Interface Software and Technology* (New Orleans, LA, USA) (UIST '19). Association for Computing Machinery, New York, NY, USA, 43–51. <https://doi.org/10.1145/3332165.3347896>
- [20] Haruki Takahashi and Homei Miyashita. 2017. Expressive Fused Deposition Modeling by Controlling Extruder Height and Extrusion Amount. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* (Denver, Colorado, USA) (CHI '17). ACM, New York, NY, USA, 5065–5074. <https://doi.org/10.1145/3025453.3025933>
- [21] Haruki Takahashi, Parinya Punpongsanon, and Jeeun Kim. 2020. Programmable Filament: Printed Filaments for Multi-material 3D Printing. *Proceedings of the 33rd Annual ACM Symposium on User Interface Software and Technology* (UIST '20) (2020), 1209–1221. <https://doi.org/10.1145/3379337.3415863>
- [22] Francesco Tamburrino, Serena Graziosi, and Monica Bordegoni. 2019. The influence of slicing parameters on the multi-material adhesion mechanisms of FDM printed parts: An exploratory study. *Virtual and Physical Prototyping* 14 (apr 2019), 316–332. <https://doi.org/10.1080/17452759.2019.1607758>
- [23] Kiril Vidimčec, Szu-Po Wang, Jonathan Ragan-Kelley, and Wojciech Matusik. 2019. OpenFab: A Programmable Pipeline for Multimaterial Fabrication. *Commun. ACM* 62, 9 (Aug. 2019), 97–105. <https://doi.org/10.1145/3344808>
- [24] Steve Wood. 2012. *Flexible Coupling*. Retrieved December 22, 2021 from <https://www.thingiverse.com/thing:38678>